An Insight-Driven Comparative Analysis of Deep Learning Models for Breast Ultrasound Image Classification

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***Abstract*—Early detection of breast tumors plays an important role in saving patients' lives. The utilization of ultrasounds in medicine is common practice since they are noninvasive and allow for analysis of dense breast tissue, although it may be hard for a person to analyze ultrasound images manually due to their high level of noise. This work aims at designing the classification framework of breast ultrasound images using deep learning techniques that can automatically detect benign/malignant cases via four differential networks, which include Custom Convolutional Neural Network, VGG16, EfficientNetB0, and DenseNet201. A total number of 9016 images was used for training and testing purposes of the four models using numerous types of image preprocessing and augmentation techniques for improving accuracy. After comparing results of each architecture, it was shown that the VGG16 model gave the highest accuracy of 98%, while DenseNet201 and Custom CNN gave comparable results, however, in case of EfficientNetB0, lower accuracy was found among the other models. Overall, the experiments proved that the pre-trained model architectures provide more accurate results than other architectures and could significantly assist doctors in faster decision making.**

***Index Terms*—Breast cancer, Ultrasound imaging, Deep learning, Convolutional Neural Network, Transfer learning, EfficientNetB0, VGG16, DenseNet201, Medical image classification, Computer-aided diagnosis**

I. Introduction

The timely diagnosis of breast cancer plays a crucial role in saving people's lives. Among all kinds of medical imaging techniques, ultrasound is often employed in clinical practice since this type of images can help to examine patients' dense breasts.However, ultrasound images might be highly noisy, low-contrast, and have uneven edges.

To solve this problem, deep learning techniques have become popular recently. CNN models and their advanced variants can efficiently extract useful information from medical images. It is known that the performance of a CNN model based on both images and segmentation could be boosted by the integration of the two types of data in an input layer [1]. CNN models with an attention mechanism and feature fusion have been developed for improving feature representation of tumors [2].Efficient and lightweight CNN architectures, including MobileNets, provide an efficient way to make an accurate classification [3].

Multi-modal learning involves data collected by means of different imaging procedures to achieve more precise diagnosis [4].Transformer models and graph-based methods have been used to deal with ultrasonic images [5], [6]. There are also other approaches which use edge information, perform segmentation, and apply multi-task learning to improve classification performance [8], [11].

Nevertheless, the issue is that we need to ensure consistent performance across different data sets. This paper seeks to develop a comparative deep learning algorithm for the classification of breast ultrasounds between the two categories, namely benign and malignant. Here, a bespoke convolutional neural network will be developed for comparison with other famous pre-trained models like VGG16, EfficientNetB0, and DenseNet201.

II. Background Review

Several deep learning algorithms have been developed for the purpose of recognizing features from the ultrasound image of the breast, specifically concentrating on feature extraction.

Taljeh et al. [1] studied the two-input deep learning architecture for ultrasound images and the corresponding segmentation masks of tumors. The reason for having two inputs is that the network can extract general and localized features using both. Thus, better feature extraction can be achieved by combining the two types of inputs, making it superior to networks with only one input type.

Asif et al. [2] In this study, the network architecture used is a two-branch network that is designed with the attention mechanism built into it. This makes the network able to attend to important tumor regions, thus helping achieve better classification of malignant and benign tumors.

Kaushik and Sharma [3] The basic idea behind this method is the development of an architecture based on MobileNetV3 in order to design a feasible breast cancer detector. The main aim would be to have an efficient algorithm.

Rizwana et al. [4]This reaearch use of multiple modalities in the learning of ultrasound images is investigated. The combination leads to generating complementary data which will aid in the accuracy of cancer diagnosis. Therefore, it is recommended that different inputs are used for achieving results.

Sivakumar and Kumari [5] This study uses transformers in the classification of breast ultrasound images. Transformer has the capability of modelling the long-term dependency relation in the visual information. The reason for the choice of transformers in this study is that feature learning becomes easier in modeling the global dependency relations.

Trang et al. [6] Proposed a graph-based convolutional network that is integrated with superpixel segmentation. Graphs can easily be used to capture the spatial correlation of image patches, therefore making it easier to learn the structural dependency of the ultrasound images.

Zhang et al. [7] An approach for segment-based attention approach is proposed in the classification of videos collected using ultrasound technology. The method is mainly focused on choosing the temporal segments of video frames and less on the selection of manual frames.

Xu et al. [8] The classification performance of the algorithm was further enhanced using the information regarding tumor edge detection. Due to edge consideration, differentiation among different tumor types can be done easily. In this way, the system makes use of the clinical information to enhance its performance.

Mo et al. [9] a neural network structure that uses a transformer which is free from region of interest specification. The neural network takes into account all parts of an image while analyzing it to get more accurate diagnoses and enhanced features.

Islam et al. [10] designed a transfer learning based model along with interpolation. Transfer learning enhances feature extraction capabilities. Interpolation increases accuracy through better representation of data in the form of more images. As a result, the proposed system performs well.

Xu et al. [11] The development of a framework with multitask learning technique enables both segmentation and classification tasks to be performed simultaneously. The multitask framework makes it easy to extract features and avoids redundancy.

Misra et al. [12] developed a new technique for bi-modal transfer learn- ing, which uses both B-mode ultrasonography imaging and strain elastography imaging. Model incorporates useful information from both techniques, which helps improve the feature learning process, thereby improving the classification accuracy.

Ding et al. [13] proposed an interactive learning technique that is not only able to detect but also classify tumors. Multimodality learning facilitates accurate feature learning. Hence, tumor localization becomes easier. Such methods prove helpful in making decisions.

Xing et al. [14] incorporated clinical knowledge into their deep learning model through BI-RADS. Integration of this knowledge makes models more interpretable. In turn, this helps improve classification accuracy.

Hamdy et al. [15] studied multimodal fusion technique through the use of ultrasound and mammogram data. Combination of different modality-based images provides complementary features.

III. Dataset Description

Experiments in this paper will be conducted using Ultrasound Breast Images database for cancer [36]. This data set is extensively used in academic researches to assess automated classification system using ultrasonography. Ultrasonography is very popular in practice since it provides an opportunity to observe internal organs in real-time without any radiation at a low cost.

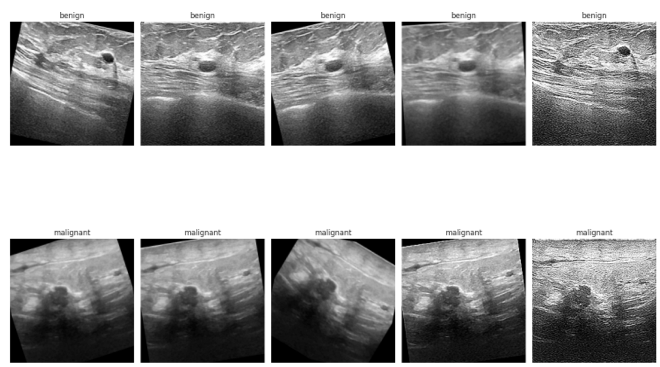

Fig. 1. Sample ultrasound images from the dataset highlight benign cases in the upper section and malignant cases in the lower section.

Examples of images from the dataset are depicted in Fig. 1 and the visual differences between benign and malignant tumors are evident. The total dataset comprises 9016 ultrasound images, and these are divided into two categories based on whether the tumor portrayed is benign or malignant. 4574 ultrasound images are of benign tumors while 4442 correspond to malignant tumors. Therefore, there is an equal number of these categories, and as such the dataset is suitable for binary classification applications. Also, the images are categorized based on their classes. Among some of the challenges posed by the images are noise artifacts, insufficient contrast as well as irregular tumor boundaries. Mathematically, the dataset is formulated as follows:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

here $x_i \in \mathbb{R}^{H \times W \times C}$ stands for the input ultrasound image while $y_i \in \{0, 1\}$ is its corresponding class label (0 is associated with benign while 1 is for malignant images).

Therefore, in this case N = 9016. The dataset is also divided into two sets for training and testing whereby one will be used for training and the other for testing purposes. As a result, about 80% of the images are for training and 20% are for testing. The allocation of images based on the two disease categories is shown in Table I.

TABLE I
BREAKDOWN OF SAMPLES BY CLASS

Category	Image Count
Benign	4574
Malignant	4442
Total	**9016**

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed model in this study aims at performing the binary classification task using breast ultrasound images which are classified into two classes, namely benignant and malignant. The set of data used in the development of the models comprises labeled images of tumors with varying features, that is, boundary, texture, and appearance. Some preprocessing methods such as resizing and normalization of images take place prior to the training step.

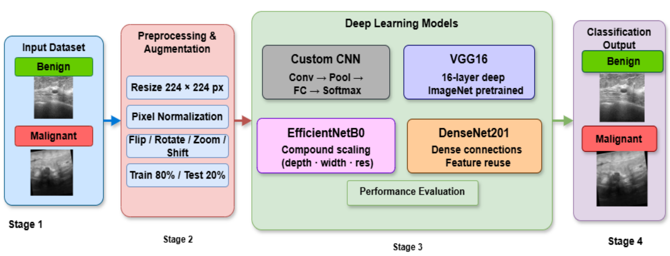

Fig. 2. Workflow of the proposed framework for analyzing breast ultrasound images.

An illustration is shown in Fig. 2 below and the architecture is broken down into three steps, namely preprocessing, feature extraction, and prediction. In addition, various types of deep learning architectures have been employed to extract the discriminative features from the input ultrasound images and classify them into their respective class labels, which include Custom CNN, VGG16, EfficientNetB0, and DenseNet201. Evaluation of the models' performances has been done using some standard metrics like precision, recall, and confusion matrix.

V. METHODOLOGY

Proposed framework concentrates on the automated identification of breast cancer from the ultrasounds. This process includes data preprocessing, augmentation, features learning, and prediction. The entire process flow is illustrated in Fig. 3. Input images were pre-processed to make sure that all of the images have the same size and quality. In order to diversify the dataset, transformations like flipping and rotating were carried out. Models use certain patterns and depend on these patterns to discriminate between benign and malignant tumors.

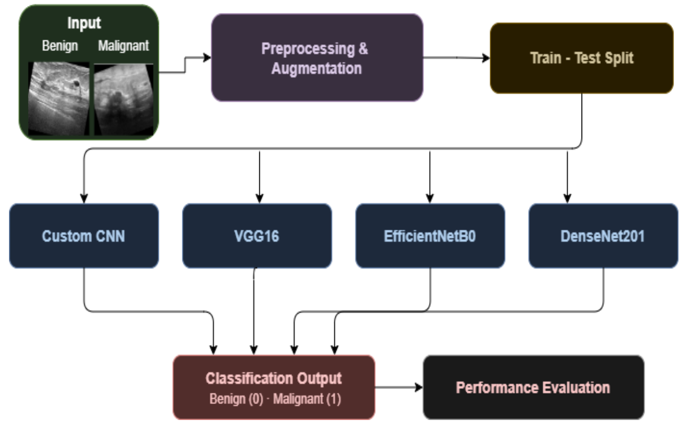

Fig. 3. Proposed Framework Architecture for Breast Cancer Classification

Fig. 3 illustrates the process of data processing implemented in the proposed approach to detection of breast cancer. The proposed process starts with the acquisition of ultrasound images, which are then pre-processed and augmented. Several models like EfficientNetB0, VGG16, DenseNet201, and a custom CNN were trained and evaluated for image classification.

Different stages are involved in the methodology of the experiment, starting with data acquisition, then preprocessing, modeling, and evaluation. Various machine learning models were tested and compared to assess their suitability for classification of breast ultrasounds.

A. Data Collection

This research employs the use of ultrasound breast image data, which consists of cases categorized into two classes namely malignant and benign tumors. This kind of data has issues such as noise, low contrast, and poorly defined tumors. Mathematically, the given data set can be represented in the following manner:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{2}$$

where x_i is the input picture, and $y_i \in \{0, 1\}$ represents its classification. The data set is used to implement an automatic mechanism for the classification of breast cancer through supervised learning methods.

B. Data Preparing

Prior to training, pre-processing of the images was done to standardize and optimize their quality, thus leading to improved performance during training. Firstly, all the images were standardized into 224 × 224 pixel size to meet the input criteria of the model.

The intensity of pixels within the images was also normalized through scaling, which leads to improved performance of the training process.

For improved reliability of the model, various transformations including rotation, zooming, flipping, and shifting were applied to different images.

C. Train-Test Split and Validation

The data set was split into two parts: training and testing, in the proportion 80:20:

$$|D_{train}| = 0.8N, \quad |D_{test}| = 0.2N \tag{3}$$

The training was performed using the training data set, while evaluation was done using the testing data set. Besides that, the method of validation was employed during training. The quality of the model was estimated using common criteria for evaluation.

D. Deep Learning Models

For the evaluation of efficiency of classification algorithms, four different deep learning algorithms have been considered.

1) Custom CNN: The convolutional neural network has been designed with convolution layer, activation function, pooling layer, and fully-connected layers for the purpose of feature extraction and classification tasks.

$$S(i,j) = \sum_m \sum_n X(i-m, j-n) K(m,n) \tag{4}$$

$$f(x) = \max(0, x) \tag{5}$$

$$P(i,j) = \max_{(m,n)\in\Omega} X(i+m, j+n) \tag{6}$$

$$z = Wx + b \tag{7}$$

2) VGG16: The VGG16 algorithm is one kind of convolutional neural network which includes the deep architecture, wherein the convolution filters with a small dimensionality are included in numerous layers. The pre-trained weight of the model was fine-tuned in this particular case.

$$\theta_{new} = \theta_{pretrained} + \Delta\theta \tag{8}$$

3) EfficientNetB0: The model of EfficientNetB0 is an upgraded version of the network parameters through optimization of depth, width, and resolution of the network at once.

$$depth = \alpha^{\phi}, \quad width = \beta^{\phi}, \quad resolution = \gamma^{\phi} \tag{9}$$

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{10}$$

4) DenseNet201: In the DenseNet201 model, the layers are interconnected with each other; thus, efficient interaction takes place between the layers.

$$x_l = H_l([x_0, x_1, ..., x_{l-1}]) \tag{11}$$

These algorithms were compared for determining which one was the most suitable for classification of breast ultrasound images.

VI. Results

The experimental tests were conducted using breast ultrasound images from the above-mentioned dataset. In this current study, two categories of images have been considered: *benign* and *malignant*. The images were resized to a size of 224×224. Then, the images were split into a training and test set at a rate of 80/20 percent.

To improve the generalization capability of the model, we performed augmentation such as image rotation, zooming, and horizontal flipping during training. Four deep neural network architectures were employed for image classification, namely, Custom CNN, VGG16, EfficientNetB0, and DenseNet201.

A. Performance Measures

The evaluation measures were calculated using the elements of the confusion matrix.

$$\begin{aligned} Accuracy &= \frac{TN+TP}{TP+TN+FP+FN}, \\ Precision &= \frac{TP}{FP+TP}, \\ Recall &= \frac{TP}{FN+TP}, \\ F1 &= \frac{2 \times Precision \times Recall}{Recall + Precision} \end{aligned} \tag{12}$$

B. Model Comparison

The deep learning models were analyzed through comparison, using commonly used evaluation measures. The results achieved by all the models were consistent and reliable. Among the models used, VGG16 was found to give the best prediction results, which proved that VGG16 learned meaningful patterns from the breast ultrasound images.

TABLE II
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN	0.96	0.95	0.96	0.95
VGG16	**0.98**	**0.98**	**0.97**	**0.97**
EfficientNetB0	0.92	0.91	0.92	0.91
DenseNet201	0.97	0.96	0.97	0.96

From all the models tested, the model which exhibited the best results was VGG16, registering an accuracy of 98%. The other models that scored well were DenseNet201 and Custom CNN. On the other hand, the model which did not perform well compared to the rest was EfficientNetB0.

C. Discussion

The results demonstrate that models based on the concept of transfer learning outperform Custom CNN for breast ultrasound classification tasks. Specifically, among these, the model based on the VGG16 framework demonstrates excellent performance, while DenseNet201 delivers comparable performance, attributable to its high level feature reuse capabilities. The model based on the Custom CNN provides reasonable results, though it did not have any pre-trained knowledge; meanwhile, EfficientNetB0 delivered less accurate results, which may be associated with its poor fine-tuning capabilities.

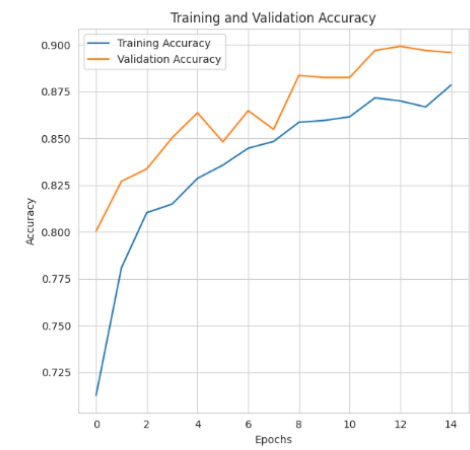

Fig. 4. Accuracy results of VGG16 during training and validation for 15 epochs, showing consistent performance with an accuracy rate of around 98%.

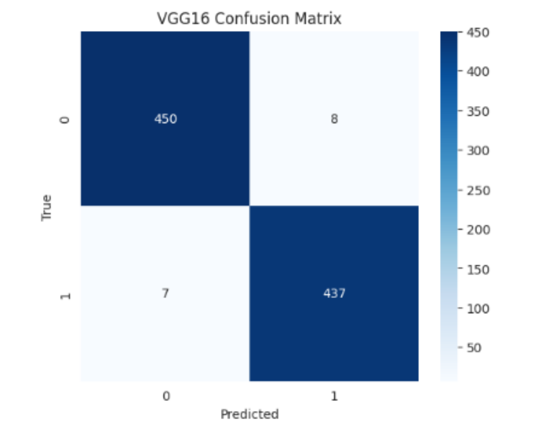

Fig. 5. Confusion Matrix of the VGG16 shows that most data are correctly classified with 98% accuracy.

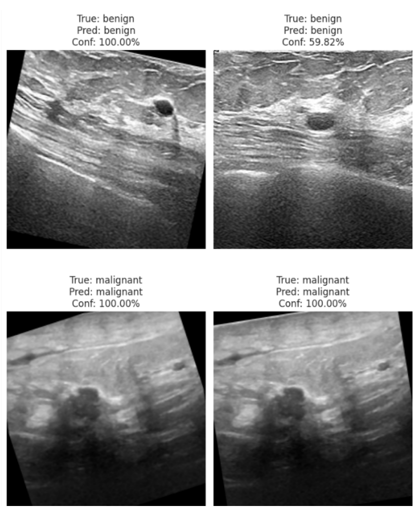

Fig. 6. Prediction results of the VGG16 model on breast ultrasound images. The figure shows both benign and malignant cases along with their predicted labels and confidence scores.

In this research, we selected models that represent a wide range of approaches, including the baselines as well as more advanced models. Specifically, Custom CNN is a reference model, while VGG16, DenseNet201, and EfficientNetB0 belong to transfer learning models.

The model based on the VGG16 framework provides excellent results as compared with other models since it fea- tures a simple uniform structure that enables fine-grained feature extraction. Pretrained weights ensure high performance in terms of generalization. On the other hand, EfficientNetB0 requires further fine-tuning, whereas Densenet201 features a high level of complexity.

VII. Conclusion

In this research, the effectiveness of applying deep learning algorithms for classification of breast ultrasounds using a publicly available data set has been investigated. As per the above discussion, it is clear that the VGG16 algorithm showed maximum efficiency of 98%, while the DenseNet201, Custom CNN, and EfficientNetB0 models exhibited efficiencies of 97%, 96%, and 92%, respectively. Based on the above analysis,

It is clear that deep learning techniques have proven to be quite effective in the process of feature extraction from the ultrasounds. Further research may be done with larger datasets, parameter tuning, and XAI.

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